

Software Requirements Specification

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Project Name:

**Information System for Managing the Activity of an Art
Enthusiasts Club**

Version 1.13

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Revision History

Name	Date	Reason For Changes	Version
Draghici Alexandru	26.02.2026	Added the Introduction section and defined the purpose of the system	v1.0
Dobre Lorena	26.02.2026	Completed Product Scope and added reference sources for similar systems	v1.1
Gherasie Gabriel	27.02.2026	Added user classes and characteristics and clarified the role of each user type	v1.2
Draghici Alexandru	01.03.2026	Added first functional requirements related to user registration, login and account management	v1.3
Gherasie Gabriel	01.03.2026	Added resource management requirements and reservation conflict rules	v1.4
Dobre Lorena	03.03.2026	Added event creation requirements and invitation management functionality	v1.5
Draghici Alexandru	04.03.2026	Extended reservation rules and added requirements related to resource availability intervals	v1.6
Gherasie Gabriel	05.03.2026	Updated event and invitation requirements and clarified event participation conditions	v1.7
Gherasie Gabriel	06.03.2026	Added payment rules and financial constraints related to membership and reservations	v1.8
Draghici Alexandru	08.03.2026	Added reporting requirements for resources, payments and monthly financial data	v1.9

Dobre Lorena	08.03.2026	Updated several functional requirements and clarified reservation conditions	v1.10
Gherasie Gabriel	10.03.2026	Added non-functional requirements including performance, software quality attributes and business rules	v1.11
Dobre Lorena	11.03.2026	Reviewed functional requirements and updated document structure before submission	v1.12
All members of the team	13.3.2026	Reviewed and updated the entire document	v1.13

1. Introduction

1.1 Purpose

The purpose of this document is to describe in detail the functional and non-functional requirements of an information system for managing the activity of an art club. The system will handle the administration of members, membership fees, physical resources (conference rooms, exhibition halls) and events organized by the club.

1.2 Document Conventions

This document follows the IEEE Software Requirements Specification (SRS) structure. Requirements are identified using the prefix REQ followed by a unique number. Functional requirements are written using the phrase “The system shall...” to ensure clarity and testability. Section titles and numbering follow a hierarchical structure. The priority is defined at the system feature level, and detailed requirements inherit the priority of their corresponding feature.

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- ❖ development team members (for implementation and testing);
- ❖ the supervising instructor (for project evaluation);
- ❖ future system administrators (to understand the capabilities of the application).

1.4 Product Scope

The product is a web application (information system) that supports managing the activities of an art enthusiasts club, with the following main capabilities:

- management of members and their membership status;
- management of payments (fees, reservation charges for non-members);
- tracking and reservation of physical resources (rooms, exhibition spaces);
- organizing events (exhibitions, workshops, conferences) and sending invitations to members.

The product will be accessed via a web browser and will be implemented using ASP.NET Core with a Microsoft SQL Server database.

1.5 References

[R1] Karl E. Wiegers, “Software Requirements Specification Template,” Version 1.0, 1999.

[R2] “Arta Inramarii” – example art exhibition site, <https://www.arta-inramarii.ro>

[R3] “Teamwork Arts” – example event-organizing site, <https://teamworkarts.com>.

2. Overall Description

2.1 Product Perspective

The system is a new, independent product that does not replace an existing system. It will be used by a single art club, but can be expanded to multiple clubs later.

2.2 Product Functions

At a high level, the product offers the following function groups:

- ❖ User and member management: registration, authentication, role definition (admin, member, external user), member profile administration.
- ❖ Resource management: creating and updating resources (rooms), managing availability, reservations, and viewing a reservation calendar.
- ❖ Event and invitation management: creating and editing events, selecting invited members, managing invitation inbox and notifications.
- ❖ Payment and reporting management: recording membership fees and reservation charges, computing revenues and expenses, generating financial reports.

2.3 User Classes and Characteristics

The following user classes are identified:

- ❖ Administrator (Admin)
 - Has full rights to manage members, resources, events and reports.
 - Can define membership fees and can make reservations at any time, as long as the resource is free.
- ❖ Member
 - Registered user with an active membership.

- Can create and edit own events, invite other members, view and manage received invitations, and reserve resources under certain conditions (financial rules).
- ❖ External User (Non-member)
 - Can reserve resources without being a member, at a higher fee (standard non-member rate).
 - Has limited access only to resource reservation and the related payment.

2.4 Operating Environment

- ❖ Client Interface : modern web browser (Chrome,Opera,Firefox,Safari) on desktop or on mobile devices.
The backend shall be implemented using ASP.NET Core and shall run on a Windows or Linux server.
- ❖ The application shall use Microsoft SQL Server as the database management system for storing data about members, resources, events, and payments.
- ❖ The system shall be deployed as a single web application, accessible over the internet or a local network, depending on the project setup.

2.5 Design and Implementation Constraints

The backend will be implemented in C# using ASP.NET Core. ASP.NET Core Identity will be used for authentication and authorization, with roles (Admin, Member, External User). Data persistence will be implemented using Entity Framework Core with Microsoft SQL Server.

2.6 User Documentation

The following documentation will be provided:

- ❖ Administrator user guide (PDF or dedicated web page);
- ❖ Member user guide (short tutorial, in-app "Help" page);
- ❖ Error messages and tooltips in the user interface.

2.7 Assumptions and Dependencies

- ❖ A stable network infrastructure is assumed between clients and the application server.
- ❖ Users are assumed to have a modern web browser and internet access.
- ❖ The system depends on the availability of an email service for sending notifications/invitations (if real email sending is implemented).

3. External Interface Requirements

3.1 User Interfaces

The user interface will be a web application with the following general characteristics:

- ❖ Responsive design, properly displayed on both desktop and mobile devices.
- ❖ Main navigation menu with access to: "Members", "Resources", "Events", "Reports" (depending on user role).
- ❖ Forms with validation (required fields, clear error messages) for login, registration, events and reservations.

Example logical screens:

- ❖ Login and user registration page;
- ❖ Admin dashboard (summary statistics, quick access to reports);
- ❖ Member list and filter page;
- ❖ Resource reservation calendar page;
- ❖ Invitation inbox page for members.

3.2 Hardware Interfaces

The system does not interact directly with specialized hardware devices. Access is through standard devices (PCs, laptops, tablets, smartphones) connected to the internet.

3.3 Software Interfaces

The application will interact with the following software components:

- ❖ Microsoft SQL Server: The database management system for storing and retrieving data about members, resources, events, and payments. Entity Framework Core will be used as the ORM (Object-Relational Mapper) for database access.
- ❖ SMTP Server: An email server for sending invitations, notifications, and password reset tokens to users.
- ❖ ASP.NET Core Identity: The authentication and authorization framework for managing user credentials, roles, and session tokens.

No integrations with external third-party APIs are planned for the initial version.

3.4 Communications Interfaces

Communication between client and server will use HTTP/HTTPS. The application will follow a traditional web architecture with ASP.NET Core handling server-side rendering (Razor Pages or MVC) and/or RESTful API endpoints for dynamic client interactions. All sensitive data will be transmitted over secure HTTPS connections in production environments."

4. System Features

The application will communicate with the email server (SMTP) to send invitations and notifications, and with the SQL Server database to store data. No integrations with other external applications are provided.

4.1 System Feature 1 - User and Member Management

4.1.1 Description and Priority

This feature covers user registration and authentication, role differentiation (admin, member, external user), and member profile administration. Priority: High.

4.1.2 Stimulus/Response Sequences

- ❖ An unregistered user opens the registration page, fills the form and, after validation, the system creates a new account of type "member" with default status (e.g., inactive until confirmed by an admin, if desired).
- ❖ An existing user submits valid credentials in the login form; the system validates the data, creates an authenticated session using ASP.NET Core Identity, and redirects the user to the dashboard corresponding to their role.

4.1.3 Functional Requirements

Authentication & user CRUD

REQ-1: The system shall provide a registration form that allows new users to create a member account by entering mandatory information (name, email, password, and any additional required fields)

REQ-2: The system shall provide a secure login function allowing all registered users (including members and admins) to authenticate using their email and password. Upon successful login, the system shall redirect users to their personalized dashboard based on their assigned role

REQ-3: The system shall allow admins to add, edit, delete, or ban users, members, and other admins through a management interface

Membership & payments at user level

REQ-4: The system shall allow admins to view and filter past membership or resource payment records by user, date, and payment status

REQ-5: The system shall allow users to purchase memberships through a secure payment gateway. Upon successful payment confirmation, the user's account shall be upgraded to a member account, unlocking additional features such as priority resource booking and access to exclusive events

Member profile & account lifecycle

REQ-6: The system shall allow members to view their profile details, including contact information, membership type, and activity history

REQ-7: The system shall allow members to temporarily deactivate or reactivate their account through the account settings page. When deactivated, the member shall not be able to log in or make reservations, but their data shall be preserved

REQ-8: The system shall allow members to modify their personal information or update their password through a self-service interface

REQ-9: The system shall include a search function enabling admins to find members by name, email, or membership ID

REQ-10: The system shall allow an admin to view a list of registered users together with their roles and account status

REQ-11: Admins shall be able change the role of an user

REQ-12: The system shall provide a password reset mechanism based on a secure token sent to the user's registered email address

REQ-13: The system shall require users to verify their email address via an activation link before they can access member-only features

REQ-14: The system shall allow members to configure their notification preferences (e.g., enable/disable email and in-app notifications for invitations and events)

4.2 System Feature 2 – Resource and Reservation Management

4.2.1 Description and Priority

This feature manages physical resources (conference rooms, exhibition halls) and their reservation for events, verifying time conflicts and applying availability rules. Priority: High.

4.2.2 Stimulus/Response Sequences

- ❖ An admin adds a new room to the system, specifying name, capacity and description; the system marks it as "available" by default.
- ❖ A member creates an event, selects a resource and a time interval; the system checks whether the resource is free in that period (including a one-day buffer before and after) and, if there is no conflict, confirms the reservation.

4.2.3 Functional Requirements

Resource CRUD & status

REQ-15: The Admin shall be able to manage resources (can add, remove, edit description of resources) including updating their name and description

REQ-16: The system shall display the current availability status of each resource (Available, Unavailable, or Reserved) on the resource list and detail pages. The status shall automatically refresh based on new reservations or admin updates

REQ-17: The system shall allow users to view a searchable list of all available resources with key details such as capacity and location

Reservation rules

REQ-18: The system shall ensure that two reservations for the same resource do not overlap in time; any reservation request that conflicts with an existing reservation shall be rejected

REQ-19: The system shall make resources unavailable for other reservations one day before and one day after each scheduled event

REQ-20: The system shall allow admins to reserve any available resource at any time, overriding standard availability restrictions

REQ-21: The system shall allow free reservations for members only when financial conditions REQ-16 and REQ-47 are satisfied

Special resource usage

REQ-22: Admins will be able to create timed exhibition halls for everybody

Resource-centric reports

REQ-23: Members or admins are able to generate a report for the remaining available resources within a selected time interval

REQ-24: Members or admins are able to generate a report for the reservation date of a resource

REQ-25: The system shall provide a search function allowing users to find resources by name, type, or capacity

REQ-26: The system shall support the definition of extensible resource attributes to allow for category-specific technical specifications (e.g., projectors, audio-visual equipment)

REQ-27: The system shall limit each user to a configurable maximum number of active future reservations for resources (e.g., rooms or halls); additional reservations beyond this limit shall be rejected

REQ-28 : The system shall display a calendar showing all reservations for the selected resource

4.3 System Feature 3 – Event and Invitation Management

4.3.1 Description and Priority

This feature allows members to create events (exhibitions, workshops, conferences), invite other members and manage received invitations. Priority: High.

4.3.2 Stimulus/Response Sequences

- ❖ An authenticated member accesses the "Create event" page, enters event details (title, description, date, time, resource) and selects invited members; the system validates the data, reserves the resource and generates invitations.
- ❖ A member receives an invitation to an event and sees it in the invitation inbox; the member can mark the invitation as "accepted" or "declined", and the organizer sees the updated status.

4.3.3 Functional Requirements

Event CRUD

REQ-29: Members shall be able to create events provided they have not exceeded their monthly event creation limit.

REQ-30: Members shall be able to edit their own events only up to two days before the scheduled event date.

Invitations & inbox

REQ-31: Members shall be able to select users to send event invitations

REQ-32: Members shall be able to reserve resources for an event if those resources are available

REQ-33: Users shall be able to access the invitation inbox system from their profile menu

REQ-34: Users shall receive automatic email notifications when they are invited to an event by a Member or an Admin

REQ-35: Users shall receive in-app notifications when they are invited to an event by a Member or an Admin.

Event location & affiliated venues

REQ-36: The system shall allow admins to add and manage affiliated event locations for club use

REQ-37: Members shall be able choose the location event or select one of the affiliated ones if said location isn't already booked.

Content around events / art pieces

REQ-38: The system should provide a "New Pieces" section showing the latest art pieces added to the club's collection, ordered by creation or addition date.

REQ-39: The system should provide a "Popular Pieces" section showing the most viewed or most reserved art pieces within a configurable recent time interval.

REQ-40: The system should provide a detailed information page for each art piece, including title, description, creator, and associated events or exhibitions.

REQ-41: The system shall allow members to cancel their events or memberships only if no outstanding payments are due and if no active resource reservations are linked to the event or account. The system shall display confirmation prompts before completing the cancellation.

REQ-42: The invited members can view event details before accepting or declining the invitation

4.4 System Feature 4 – Payments, Revenues, Expenses and Reports

4.4.1 Description and Priority

This feature covers recording membership fee payments, charging resource reservations for non-members, computing expenses generated by events and generating financial reports. Priority: High.

4.4.2 Stimulus/Response Sequences

- ❖ An admin records a membership fee payment for a member; the system updates the member's balance and the club's revenues for the current month.
- ❖ An event that uses resources is created; the system automatically computes the associated expenses (200 lei/day/art piece , 300lei/day/location) and includes them in the current month's total.
- ❖ At the end of a month, an admin generates the income and expenses report; based on it, the system decides whether members will be allowed to make reservations in the following month (if expenses exceed revenues, member reservations are blocked, but admin reservations remain allowed).

4.4.3 Functional Requirements

Core payments & systems

REQ-43: The system should allow admins to record additional payments for resources, memberships, or events

REQ-44: The system shall allow admins to define and update payment amounts for user contributions and tariffs

REQ-45: The application shall include an integrated payment system for membership fees, resource rentals, and other related payments.

Financial rules & tariffs

REQ-46: For every event that uses at least one resource, the system shall record an expense of 200 lei per day per resource for the club.

REQ-47: If, for a given month, the total income of the club is less than the total expenses, then for the following month member reservations of resources shall be blocked, while admin reservations remain allowed.

REQ-48: For non-members who reserve resources, the system shall charge a fee of 400 lei per day per resource and record the corresponding income.

Reports (money-centric)

REQ-49: The system shall generate reports of all income received from user payments within a specified timeframe

REQ-50: The system shall generate monthly reports summarizing total income, expenses, and club balance

REQ-51: The system shall allow administrators to export financial reports in PDF format for review or archiving

4.4.4 Expected errors

- ❖ ERR-01: The system shall display a login error when the email/username or password is incorrect.
- ❖ ERR-02: The system shall display a registration error if an email address is already associated with another account.
- ❖ ERR-03: The system shall display a payment error when a payment transaction cannot be completed.
- ❖ ERR-04: The system shall display a reservation error if the selected resource (piece or room) is already reserved.
- ❖ ERR-05: The system shall display a "Member not found" error when a user attempts to send an invitation to a non-existent user.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The system shall respond in under **5 seconds** for common operations (login, listing members, listing resources) with up to **20 concurrent users**.

Generation of monthly income and expenses reports shall not exceed **10 seconds for up to 10,000 payment records**.

5.2 Safety Requirements

The system does not involve safety-critical operations or interactions that could result in loss, damage, or harm, since it is a web-based information system intended only for administrative and organizational use. Consequently, no safety safeguards, regulations, or certifications are applicable.

5.3 Security Requirements

SEC-01: The system shall implement ASP.NET Core Identity for user authentication, supporting secure password hashing, email verification, and role-based authorization for users, members, and admins.

SEC-02: The system shall enforce secure password policies, requiring minimum length (12 characters), complexity (uppercase, lowercase, numbers, symbols), and preventing common weak passwords.

SEC-03: The system shall implement session management with configurable expiration (default: 30 minutes of inactivity) and automatic logout after prolonged inactivity or suspicious activity.

SEC-04: The system shall use HTTPS for all communications and store sensitive data (passwords, tokens, payment info) encrypted at rest using industry-standard algorithms.

SEC-05: The system shall provide protection against common web vulnerabilities including CSRF, XSS, SQL injection, and brute-force login attempts (rate limiting: max 5 attempts per hour per IP).

SEC-06: The system shall log all security-related events (failed logins, privilege escalations, payment attempts) with timestamps, user IDs, and IP addresses for audit trails.

SEC-07: The system shall generate and validate secure tokens for password resets and email verification, with automatic expiration (24 hours) and single-use enforcement.

5.4 Software Quality Attributes

- ❖ **Usability:** The interface shall be simple and intuitive, with clear forms and error messages, suitable for non-technical users.
- ❖ **Maintainability:** The codebase shall be organized into modules (users, resources, events, payments, reports) to ease future modifications and extensions.
- ❖ **Reliability:** The system should operate without critical errors at least 99% of the time under normal usage conditions.
- ❖ **Portability:** The backend solution shall be implemented using ASP.NET Core, which supports deployment on multiple operating systems (Linux, Windows) with Microsoft SQL Server or compatible database systems.
- ❖ **Single Responsibility Principle (SRP):** Each class shall have only one reason to change, with clear separation of concerns (e.g., Controllers handle HTTP requests, Repositories handle data access).
- ❖ **Open/Closed Principle (OCP):** Software entities (classes, modules, functions) shall be open for extension but closed for modification, achieved through dependency injection and interfaces (e.g., IReservationService interface for different reservation strategies).

- ❖ **Liskov Substitution Principle (LSP):** Subclasses shall be substitutable for their base classes without altering the correctness of the program (e.g., different types of resources inheriting from a base Resource class).
- ❖ **Interface Segregation Principle (ISP):** Clients shall not be forced to depend on interfaces they do not use.
- ❖ **Dependency Inversion Principle (DIP):** High-level modules shall not depend on low-level modules; both shall depend on abstractions (e.g., Services depend on repository interfaces, not concrete implementations).
- ❖ **DRY (Don't Repeat Yourself):** Business logic shall not be duplicated across the application; shared functionality (e.g., payment validation, reservation conflict checking) shall be centralized in reusable services or helper methods.
- ❖ **Object-Oriented Programming (OOP):** The application shall follow OOP principles including encapsulation (data hiding in private fields with public properties), inheritance (base classes for related entities), and polymorphism (interfaces for varying behaviors).
- ❖ **Repository Pattern:** Data access shall be abstracted through repository interfaces and implementations, providing a clean separation between business logic and data persistence.
- ❖ **Dependency Injection (DI):** All service dependencies shall be injected through constructor parameters using ASP.NET , promoting testability and loose coupling.
- ❖ **Unit of Work Pattern:** Complex transactions involving multiple repositories (e.g., creating an event with resource reservation and invitations) shall be coordinated through a Unit of Work pattern to ensure data consistency.

5.5 Business Rules

- ❖ Only admins can define membership fees and modify user roles.
- ❖ Only members with an active membership can create events and invite other members.
- ❖ External users can only book resources at the non-member rate and cannot access member-only features.

6. Other Requirements

No additional requirements are currently identified beyond those described in the previous sections. Future versions of the system may introduce additional requirements depending on the evolution of the project.

Appendix A: Glossary

- ❖ Admin – User with full management privileges in the system.
- ❖ Member – Registered user with an active membership.
- ❖ External User – User without membership who can reserve resources for a higher fee.

- ❖ Resource – A room or exhibition space that can be reserved.
- ❖ Event – An activity organized by members of the club.
- ❖ SRS – Software Requirements Specification.

Appendix B: Analysis Models

Analysis models are not included in the current version of this document.

In future versions, the system will include several diagrams used for system analysis and design, such as Use Case diagrams, Activity diagrams, Sequence diagrams, Class diagrams, and Entity-Relationship (ER) diagrams.

Appendix C: To Be Determined List

No items are currently marked as TBD in this document.